

ARNOB RIZWAN AHMAD

AI / LLM Engineer • Agentic Systems • RAG & Evaluation • Python Backend • Cloud

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PROFILE

AI/LLM engineer and Software Engineering graduate of Universiti Teknologi Malaysia (July 2026), moving into AI from production full-stack and ERP delivery. Build LLM systems with that same operational discipline — agentic workflows, RAG over FAISS and pgvector, and evaluation harnesses with blinded LLM-judge scoring and execution guardrails, on the FastAPI, PostgreSQL and Kubernetes infrastructure they run on. Certified AMD ROCm Associate. Seeking AI/ML engineering roles where evaluation and reliability matter as much as model choice.

EDUCATION

Universiti Teknologi Malaysia (UTM) — Johor Bahru

Oct 2022 – July 2026

Bachelor of Software Engineering (Computer Science) — Graduated July 2026

PROFESSIONAL EXPERIENCE

Enterprise Resource Planning Developer — Hirdaramani Bangladesh — Hybrid (Full-time)

Jul 2026 – Present

ERP systems for a multinational apparel manufacturing group

- Build and maintain ERP modules supporting production, inventory and order-management workflows across factory operations.
- Write and tune SQL against large operational datasets; build reports and data integrations between ERP and internal systems.
- Work with operations and IT stakeholders to translate process requirements into shipped features.

Agentic AI Engineer — BLOOMTECH, Inc. — Remote (Part-time)

Aug 2026 – Oct 2026

Agentic AI systems and LLM-driven tooling

- Built agentic AI workflows in Python — multi-step LLM planning, tool and API integration, and structured output handling.
- Designed prompt and evaluation loops that measured and improved agent reliability across task categories.

Application Developer (Freelance) — Moien Delivery — Luxembourg (Remote)

Mar 2026 – Jul 2026

Hybrid logistics platform serving customers, restaurants and drivers — live on Google Play & Apple App Store

- Built the Angular web front-end and a cross-platform React Native mobile app on a Nuxt.js backend, delivering features for three distinct user roles.
- Implemented real-time order tracking, driver dispatch and delivery workflow automation over REST APIs and WebSocket event streams.
- Owned features from spec through QA to production deployment on weekly release cycles with international stakeholders.

Machine Learning Engineer (Contract via Mercor) — AI News HUD — USA (Remote)

Oct 2025 – Mar 2026

US-based AI-powered news intelligence platform

- Built and evaluated production PyTorch models for real-time news classification, summarization and personalized recommendations.
- Designed evaluation pipelines and ran iterative experiments on live traffic, raising news-classification accuracy from **70%** to **78.9%** and cutting inference latency **20%**.
- Took ML components from prototype to deployed feature in the platform's core intelligence layer.

Freelance & Contract Engineering

Sep 2024 – Present

Concurrent client and programme work delivered alongside study

- **PI STAR** (Bangladesh, Sep 2026) — resolved a total production outage on a Node.js and Express storefront running under Phusion Passenger on cPanel. Every URL returned 503, including static files, so the fault sat in the account and not the app. The process was dying on boot and respawning every two seconds, which held the CloudLinux cage at 100% CPU. I cleared orphaned installs that had exhausted the fork limit, then traced the silent boot failure to a prebuilt better-sqlite3 binary segfaulting before any JavaScript error could throw, and pinned a self-compiling build to fix it.
- **FF Gamer BD** (Bangladesh, Sep 2026) — backend integration on a live Laravel 10 and Filament game top-up storefront. Wired in a reseller API as an automated fulfilment provider, then built a payment-gateway integration whose webhook re-verifies every callback against the provider before an order is marked paid. Both came down to the same fault: the checkout path bypassed the storefront's own provider abstraction with hardcoded calls. I audited the product-to-SKU mapping before switching automation on, which caught a duplicate identifier that would have mis-delivered an entire product line.
- **Framic** (remote, Aug 2025 – Jul 2026, ~10 hrs/week) — founding backend engineer at an AI storage startup; designed backend services from scratch, owned architecture and core API design, and led AI search and real-time file indexing.
- **MOMS Mobile Oil Change** (USA, Apr – Jul 2026) — production Next.js site, live at momsoilchange.com: rebuilt navigation and three product pages, led an SEO overhaul, and ran a site-wide security and dependency audit.
- **Headstarter** (USA, Sep 2024 – Mar 2025) — competitive 6-month software engineering residency; shipped ML and full-stack projects end to end, and cut image size and build time on a Docker-based GitHub Actions pipeline.

SELECTED PROJECTS

Support Welfare Companies — Worker-Welfare Rating App (Google Play)

Sep 2026

- Shipped a Flutter Android app, live on Google Play, that lets the public rate employers across five worker-welfare criteria, paired with a React and TypeScript reviewer panel that checks each submission against public evidence before it scores.
- Modelled a submission as a claim to be verified rather than a vote to be counted, and blocked ballot-stuffing with Firestore rules that gate every read and write behind an admin check plus a disposable-email blocklist.
- Built with a collaborator on Firebase (Firestore, Auth, AdMob). Source repo private.

Ambient Ops Triage Workbench — Data Forensics & Retool Operations Tool

Sep 2026

- Handed a quarterly quality memo and a replica schema, found the memo's SQL was correct and its answer was not: rows were one per ingestion rather than one per note, a type-2 employee dimension fanned out every join, two different people shared a name and were silently merged by a GROUP BY, and an effective-dated rubric and SLA both changed on the same day, moving the pass rate in opposite directions.
- Showed that 149 apparently open escalations were 43 rows stranded by a rate-limited batch job with no retry.
- Built the keyboard-only triage workbench on Retool over Postgres: commits a decision in three keystrokes, restores selection, filter, sort and scroll across a refresh, and cold-loads in under ten seconds off a materialised view.

Relay — Real-Time Chat, Offline-First (taghyeer-chat-rust.vercel.app)

Sep 2026

- Built real-time direct and group chat on Next.js and TypeScript with an offline outbox that survives a reload, normalising every wire shape at one network boundary so no component ever sees the raw API.
- Sent messages over REST rather than the socket because the socket ack carried no body to reconcile against, and kept tabs signed in as the same user in sync over BroadcastChannel and navigator.locks.
- Designed for a free-tier backend that sleeps: the landing page starts waking it while you read, and a slow request shows a counter rather than an unexplained spinner. 91 unit tests, clean lint and type check.

Al-Ibadah — Bangla Islamic Reference Platform (alibadahfoundation.com)

Sep 2026

- Shipped a Bangla-first Islamic reference site, live at alibadahfoundation.com: the full Quran with five Bangla translations, three tafsirs and ten reciters, 36,432 hadith across 8 collections, duas, and prayer times and qibla for all 64 districts.
- Built an in-browser question-answering assistant over a hand-written, source-cited corpus — BM25 plus character-trigram retrieval with a confidence threshold that refuses rather than fabricates, needing no model or API key at runtime.
- React 18, Vite and TypeScript exported to a fully static build with all 141 routes prerendered for SEO, deployed on GitHub Pages with a custom domain and HTTPS.

School Care App — 40-School Deployment — Sylhet, Bangladesh

Mar 2026

- Delivered a full-stack school management system adopted by 40 schools across Sylhet; in closed testing ahead of Google Play launch.
- Built a React admin dashboard and a React Native app for parents and teachers, with Cloud Functions fanning out role-targeted FCM push notifications by school, role, and class.

ShopLocalr — Local Commerce Mobile App — Malaysia

Feb 2026

- Shipped a Flutter local-business discovery app with customer, business, and admin roles — QR deal redemption with checkout PINs, Gemini-backed recommendations, and offline-first Hive caching on a Firebase backend.

Autonomous ML Agent — LLM-Planned AutoML Pipeline (GitHub)

- Built an AutoML system where an LLM planner proposes modeling strategies under a time budget, Optuna tunes XGBoost/LightGBM/scikit-learn models, ensembles are built, and SHAP explains the winner.
- Shipped with model cards, a FastAPI inference service, an automated test suite, and CI/CD publishing to Docker Hub.

Self-Improving Prompt Optimization API — LLM Evals Infrastructure (GitHub)

- Built CI/CD for prompts: a semver-versioned prompt registry where each improvement cycle runs failure analysis, generates candidates, A/B-evaluates them with a blinded LLM judge, and auto-promotes winners per YAML rules.
- Multi-dimensional scoring across OpenAI and Anthropic providers with FastAPI, Postgres, and Redis.

CrisisCohort — Multi-Agent AI Climate Resilience Platform (GitHub)

- Architected a climate-resilience platform for smallholder farmers with 10+ specialized AWS Bedrock agents (water optimization, climate migration, grant matching, disease diagnostics) deployed on AWS Lambda.
- Designed an AWS architecture spanning Lambda, Bedrock, DynamoDB, CloudFront, API Gateway, and Cognito, plus an on-chain water-rights trading contract in Solidity.

Terminal Coding Agent — Sandboxed CLI Agent (GitHub)

- Built a pip-installable CLI coding agent that plans edits via OpenAI/Anthropic, applies them through a diff manager, and executes inside E2B Firecracker microVMs.
- Added guardrails that scan for leaked credentials (AWS, GitHub, Slack) before any edit is applied, covered by an automated test suite.

TECHNICAL SKILLS

ML / AI: PyTorch, scikit-learn, XGBoost, LightGBM, Optuna, SHAP; LLM agents, RAG (FAISS, pgvector), LLM evaluation & blinded judges, prompt engineering, AWS Bedrock, Gemini

GPU / ROCm: AMD ROCm stack and HIP programming model (ROCm Certified Associate, 2026)

Languages: Python, SQL, TypeScript, JavaScript, C#, Java, PHP, Dart

Backend: FastAPI, Django + DRF, Node.js, Express, NestJS, ASP.NET Core, Laravel, Celery, REST APIs, WebSockets

Frontend & Mobile: React, Next.js, Angular, React Native, Flutter

Cloud & Data: AWS, Microsoft Azure, GCP, Docker, Kubernetes, Linux, CI/CD (GitHub Actions, Jenkins); PostgreSQL, MySQL, SQL Server, MongoDB, Redis, Retool, ERP modules and SQL tuning on large operational datasets

CERTIFICATIONS

- **AMD — ROCm Certified Associate** (Aug 2026)
- **AWS** — 4 certificates in LLMs on AWS, SageMaker JumpStart and Prompt Engineering (Mar 2025)
- **IBM** — Full Stack Software Developer Professional Certificate (Aug 2024)
- **Microsoft Azure AI** — Azure OpenAI Service, Natural Language Processing, Document Intelligence (Jun 2024)